Practical 9: Mafic and silicic intrusions

On successful completion of this practical, you should be able to:

- Recognise and describe modal layering in mafic igneous rocks
- Predict the extinction angle of plagioclase in thin section using information about its composition
- Recognise and describe the mineralogy and textures of granites
- Interpret the formation mechanism and tectonic origin of a granite, based on its mineralogical characteristics

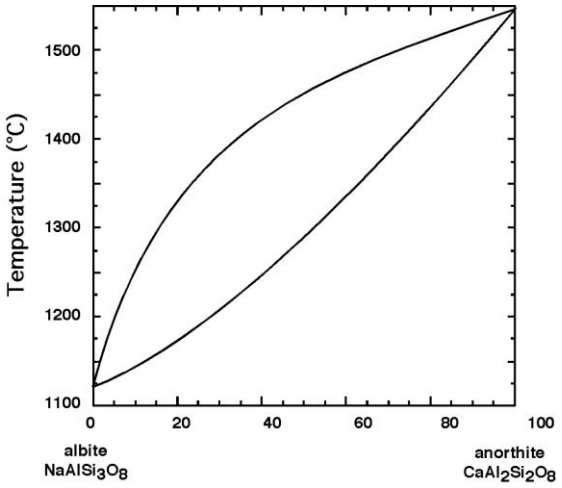
You will also practice the following skills:

- Interpret the order of crystallization in complex igneous rocks from observed igneous textures

Part 1 (45 minutes): Layered cumulate gabbros from the Skaergaard Intrusion

Minerals in the cumulate layers in the Skaergaard Intrusion change composition with increasing height in the intrusion. This reflects the continuous chemical evolution of the melt as temperature decreases and earlier-crystallizing minerals settle out.

Plagioclase: The albite-anorthite phase diagram shows how temperature affects the composition of crystallising plagioclase.



What is the high-temperature crystallizing endmember of the plagioclase solid solution? What happens to the composition of plagioclase crystallising from a cooling magma?

The high-temperature crystallizing endmember is anorthite (An). As crystallization progresses, the composition of crystallizing plagioclase becomes more albitic (more Na-rich) and less anorthitic (less Ca-rich).

Plagioclase compositions can be measured in a thin section because the extinction angle varies systematically with composition. The procedure for using extinction angles to determine plagioclase composition is explained in the Reference Handout (p 24).

Olivine: The maximum interference colour of olivine varies with composition: more Fe-rich (fayalitic) the olivine, the higher the birefringence. This can be challenging to observe in a thin section, because the interference colour is also dependent on the orientation of the crystal in the thin section. Therefore, it is only the maximum interference colour that provides information on composition.

The maximum birefringence of forsterite is 0.035. The maximum birefringence of fayalite is 0.052. What interference colours do these values of birefringence correspond to? You will need to refer to a Michel-Levy chart.

Forsterite: maximum birefringence = 0.035, corresponding to second-order sensitive tint (pink)
Fayalite: maximum birefringence = 0.052, corresponding to third-order pink

Pyroxene: Pyroxene solid solutions and crystal structures are more complex than those of olivine or plagioclase, so pyroxenes have more complex differentiation trends. In the early stages of differentiation, it is common for two pyroxenes to crystallise: a Ca-rich clinopyroxene (augite) and a Ca-poor orthopyroxene (enstatite). If the cooling rate is sufficiently slow, exsolution lamellae may develop in pyroxenes:

- Ca-rich cpx will exsolve thin lamellae of Ca-poor opx
- Ca-poor opx will exsolve thin lamellae of Ca-rich cpx

The thickness of the exsolution lamellae is proportional to the cooling rate: slower cooling produces thicker lamellae. In the case of extremely slow cooling, a texture called "inverted pigeonite" may develop, in which a host orthopyroxene crystal may contain two generations of augite exsolution lamellae in two different orientations.

Skaergaard thin sections

EG4381

What are the three principal silicate minerals in this rock? What are their modal proportions? Which are the cumulus minerals, and which are the intercumulus minerals?

The three silicate minerals are plagioclase, olivine and augite. Plagioclase and olivine are the cumulus minerals. Augite is present as both cumulus crystals and as interstitial intercumulus crystals.

Attempt to measure the composition of the plagioclase. You will need to search for a crystal in a suitable orientation: both sets of twins must be an equal shade of grey when the twin plane is oriented N-S on the cross-hairs. Do not spend too long on this: if you can't find a suitable crystal, move on to the next question.

The maximum plagioclase extinction angles are around 35°, corresponding to a composition of ~An_{60} (labradorite).

Estimate the highest interference colour of the olivine. You will need to look around the thin section and record the interference colours of several different olivine crystals before deciding on the maximum interference colour in the thin section.

The maximum interference colours of these olivines are around mid second-order (green, yellow, orange). Do not worry if you found it challenging to determine the precise interference colour.

The plagioclase has a distinctive preferred orientation, forming a lamination in the sample. This is typical of plagioclase-rich cumulates. How would you describe this texture, and how might it have formed?

This is a cumulus texture. It forms through gravity settling of crystals. The tabular shape of plagioclase means that when it settles it forms the distinctive lamination. This sample has also undergone some compaction. Evidence for compaction comes from: (1) Bent plagioclase crystals – best seen under crossed polars: some crystals have lamellar twins that are not perfectly straight; (2) Relative lack of intercumulus minerals, suggesting that liquid (melt) has been expelled from the pore spaces as the crystal pile compacted.

EG4314

This rock contains three principal silicate minerals, plus the non-silicate mineral **apatite**. Make sure you can identify the apatite. What are its optical properties in plane-polarized light and under crossed polars?

Apatite is colourless in PPL with moderate relief. It has hexagonal crystal symmetry and tends to form euhedral prismatic crystals. The basal section is approximately hexagonal in shape, and is isotropic in XPL. The prismatic section has low first-order interference colours (grey-white).

Measure the composition of the plagioclase and estimate the highest interference colour of the olivine. (Do not spend too long on this task - see comments above.)

The maximum plagioclase extinction angles are around 25°, corresponding to a composition of ~An_{40} (andesine).
The olivines have maximum interference colours around third-order green. Do not worry if you found it challenging to determine the precise interference colour. The main observation is that the olivines in EG4314 tend to have higher interference colours than those in EG4381, so they must be more Fe-rich.

Interpretation: One of the samples EG4381 and EG4314 comes from the Lower Layered Series of the Skaergaard Intrusion (500 m up), and the other comes from the Upper Layered Series (2000 m up). Assuming that the Skaergaard Intrusion crystallised from the bottom up, which is which? Explain your reasoning.

EG4381 comes from the lower layered series. EG4314 comes from the upper layered series. Explanation:
- EG4381 has more anorthitic plagioclase and more forsteritic olivine than EG4314, so it must have crystallized earlier, from a more mafic liquid. Therefore it must be from the lower part of the intrusion.
- EG4314 contains apatite, which will only crystallize at a late stage when the phosphorus concentration in the melt is high. So this rock must come from the upper part of the intrusion, near the sandwich horizon.

Even if you were not able to measure the compositions of the plagioclases and olivines in the two samples, the presence of apatite in EG4314 tells you that this gabbro must have crystallized from a more evolved melt.

Part 2 (45 minutes): Thin sections of granites

S-type Granites

The Cornubian Batholith of SW England is a composite, peraluminous, S-type granite body. Most of the rocks are biotite granites, but abundant muscovite indicates their peraluminous nature. Some of the granite compositions are extremely differentiated: they have high concentrations of boron, lithium and fluorine, so they can contain Li- and F-rich muscovite, topaz (F-rich) and tourmaline (B-rich). At some locations the hydrothermal alteration has been very intense.

14911 **or** 14910 *Lands End, Cornwall*
These thin sections are examples of a relatively undifferentiated member of the Cornubian Batholith. What are the three main colourless silicate minerals?

Alkali feldspar, plagioclase, quartz

What internal textures can you observe in the three colourless minerals? What can these textures tell you about the rock's history?

Alkali feldspar is micro-perthitic, indicating a relatively slow cooling rate. Plagioclase is zoned; this is probably normal zoning rather than oscillatory zoning and reflects the chemical evolution of the magma during crystallization. The feldspars (especially plagioclase) also show alteration to sericite, indicating some degree of hydrothermal alteration after solidification.
Quartz sometimes has undulose extinction or is beginning to recrystallize into subgrains; both these textures are evidence of deformation.

Other primary minerals: The biotite contains abundant inclusions of zircon (with pleochroic haloes), and prismatic apatite and platy ilmenite are also common. Some thin sections contain accessory topaz.

Secondary minerals: A few of the thin sections contain euhedral (square/rectangular to hexagonal) pseudomorphs after cordierite. These areas were once crystals of magmatic cordierite (an aluminosilicate mineral), but have now been completely replaced by fine-grained secondary clay minerals and sericite. Muscovite and tourmaline are common late-stage hydrothermal minerals.

I- and A-type Granites

The I-type granites from Donegal, NW Ireland, are members of the Newer Caledonian Series, intruded into Dalradian metamorphic rocks during the Caledonian Orogeny. They have radiometric ages of 390-410 Ma. The magmatism was associated with the subduction of oceanic lithosphere during closure of the Iapetus Ocean. The granitic rocks include tonalite, granodiorite and granite, and there are also some associated minor appinite intrusions (appinite is a hornblende- and plagioclase-rich plutonic rock).

The Eastern Red Hills granites on the Isle of Skye, Scotland, were emplaced late in the development of the Hebridean Tertiary igneous province at around 57-55 Ma. The magmatism was associated with the opening of the North Atlantic ocean. Tectonically and chemically, the Eastern Red Hills granites are perakaline A-type granites. However, the alkali enrichment is not very strong, and these rocks are mineralogically very similar to metaluminous I-type granites.

16265 *Ardara pluton* **or** M2 *Beinn an Dubhaich granite, Skye*
Choose either 16265 or M2 to examine in detail, but make sure you look at both as they are texturally quite different. There are only five thin sections of 16265, so please return this sample as soon as you have finished with it.

For both thin sections: What are the colourless silicate minerals? Estimate their modal proportions and use this information to classify the rock using the QAPF classification scheme.

The colourless minerals are quartz, plagioclase, and alkali feldspar. They have slightly higher modal plagioclase than the S-type granites. Your QAPF classification will depend on your estimated modal proportions, but you may arrive at a classification of either granite or granodiorite.

For both thin sections: What are the two mafic minerals, and what are their pleochroic colours? Note the absence of Al-rich minerals like muscovite.

The mafic minerals are hornblende and biotite. The hornblende pleochroic colours are dark green, mid-green and yellow. The biotite pleochroic colours are green-brown – this is characteristic of I-type granitoids, and is slightly different to the straw yellow-brown colours seen in biotites from S-type granitoids.

16265 only: Attempt to identify primary magmatic epidote and sphene crystals in this sample. Notice the strong preferred orientation of the mafic minerals. This mineral alignment was caused by the magma crystallizing under an inflationary stress field: the biotites in particular are aligned perpendicular to the direction of the principal compressive stress.

The preferred orientation is best seen holding the thin section up to the light, rather than under the microscope.

M2 only: You may find rare epidote in this sample. Describe the appearance of the feldspars in this sample and suggest a reason for their appearance.

The feldspars have a dusty texture and are extensively altered to clay minerals. This alteration is caused by water, and suggests that there was a dynamic hydrothermal system surrounding the granite. The Beinn an Dubhaich hydrothermal system probably involved meteoric water and a significant component of magmatic water.

16272 *Appinite, Ardara pluton*
Appinite occurs as minor intrusions around the margins of the Ardara pluton, as well as inclusions within the main pluton, indicating that it crystallised before the enclosing granodiorite. The rock type is named after Appin, the region of Argyll in Scotland where it was first described. Appinites are mafic plutonic rocks, so they must have formed by mantle melting (and not crustal melting). If this sample was classified using the QAPF scheme, it would probably be a hornblende diorite.

Examine the hand specimen and thin section. What are the main silicate minerals in this rock?

The most abundant minerals are green-brown pleochroic hornblende, and interstitial plagioclase, showing widespread alteration to sericite. There is also interstitial quartz and calcite. Various other minerals are present in low modal proportions, including magnetite, muscovite, biotite, apatite, epidote.

Part 3 (15 minutes): Hand specimens of granites

The hand specimens show a range of compositions and textures reflecting their different origins and evolution.

15276 *From the Strontian granite, near to Ballachulish*
What are the principal felsic minerals in this specimen? Estimate their modal proportions and use this information to classify the rock using the QAPF classification scheme.

The principal felsic minerals are plagioclase and quartz, and there is not much alkali feldspar. This is a tonalite.

9753, 9781, 9813, 9817 *"Xenolithic Ballachulish Granite"*
What are the likely origins of the xenoliths in these hand specimens?

The xenoliths are entrained fragments of the country rock. Some xenoliths are clearly foliated, so these must be meta-sediments. Some xenoliths are igneous – they are too coarse-grained to be sedimentary – and they have a range of grain sizes and colour indices. Darker coloured mafic xenoliths could represent

fragments of an early-crystallizing mafic magma that has been entrained in the granite. Other xenoliths could be mixed magma, or else they have the same bulk composition as the main granite but crystallized earlier and faster.

9805A *Contact of appinite with hornblendite, Appin*
Hornblendite is a hornblende-rich mafic to ultramafic rock. Comment on the possible origins of this hand specimen.

The most mafic mineral is hornblende, so this is likely to have crystallized first. The appinite could be a fractionation product of hornblendite, or a product of assimilation of felsic material into a more mafic magma.

Bonus thin sections: Palisades Sill

If you have finished and would like some more practice interpreting samples from mafic intrusions, take a look at these thin sections from the Palisades Sill.

11396 *Chilled margin*
When the Palisades Sill magma was intruded, a thin layer of magma at the base of the intrusion cooled and crystallised rapidly as it came into contact with the cold country rock. This chilled margin represents the initial composition of the magma, because it did not have time to differentiate before it cooled. It is a typical basalt, containing microphenocrysts of olivine, plagioclase and clinopyroxene. What features do you observe in the olivine? What minerals are in the groundmass?

The olivines are fractured and partially altered to serpentine, which suggests hydrothermal alteration. The groundmass contains plagioclase, augite and oxides.

11393 **or** 11398
This sample is much coarser-grained than the chilled margin. What are the four main silicate minerals in this rock? How would you describe the texture?

The four main silicate minerals are olivine, augite (clinopyroxene), orthopyroxene and plagioclase. The texture is poikilitic.

Notice the unusual textural relationship between the biotite and the oxides. The biotite surrounds the oxides: it is using the oxide minerals as a topotactic template, which means that crystallographic structure of the oxide is providing a very good nucleation point for the biotite.

11394
Which of the main silicate minerals in 11393/11398 is missing from this sample? Notice the exsolution textures in the pyroxenes, and the very fine-grained granophyric intergrowths of quartz and alkali feldspar.

This sample contains plagioclase, augite and orthopyroxene, but no olivine. Olivine and quartz will not co-exist in the same rock, so the presence of quartz-alkali feldspar granophyric intergrowths is consistent with an absence of olivine in the thin section.

Interpretation: Samples 11393/11398 and 11394 come from different heights in the Palisades Sill intrusion. Assuming that the sill crystallized from the bottom up, which sample comes from higher in the sill?

11394 comes from higher up in the intrusion. The evidence for this is:
- Olivine is present in 11393/11398, but not in 11394. Olivine is an early-crystallizing mineral, so the olivine-bearing samples must come from lower down the sill.
- Granophyric intergrowths of quartz and alkali feldspar must crystallize from a fairly evolved melt, which suggests that sample 11394 comes from fairly close to the sandwich horizon.

Bonus digital resources: Hand specimens, virtual outcrops and thin sections

Layered gabbro: Layered troctolite, Stokes Mountain, California

Granodiorite, Loch Doon

I-type granite: Granite, Shap
S-type granite: Biotite granite, Luxulyan
A-type granite: Aegirine granite, Rockall

A very unusual granitoid sample, just for fun: Arfvedsonite Kakortokite, Greenland. This sample comes from the Ilimaussaq intrusive complex, a layered intrusion in Greenland consisting of alkaline evolved rocks such as syenites. Ilimaussaq is known for its wide variety of rare and unusual minerals. It is the type locality for 34 minerals, including:

- Arfvedsonite, a sodic amphibole – Rotation 1;
- Eudialite, a sodic cyclosilicate used as a minor ore of zirconium – Rotation 2;
- Aenigmatite, a sodic single-chain silicate (but not a pyroxene) – Rotation 4;
- Sodalite, a sodic framework silicate – not in this rock, but widely used as an ornamental gemstone.

Take a look at the thin section and enjoy the appearance of these rare and brightly birefringent minerals. You don't need to learn these minerals or be able to recognise them.